CUSTOMER CHURN & RETENTION ANALYSIS

End-to-End Data Analytics & Machine Learning Project

PROJECT SNAPSHOT	DETAIL
Business problem	Understand customer churn patterns and prioritize retention opportunities.
Dataset	IBM Telco Customer Churn; 7,043 customer records.
Tools	Python, Pandas, NumPy, Scikit-learn, Google Colab and Power BI.
Model	Class-weighted Random Forest classifier.
Decision threshold	0.55, selected using cross-validation predictions.

What this report provides

A clear explanation of the business problem, dataset, data preparation, model workflow, evaluation metrics, risk segments, dashboard design, business recommendations, and limitations.

Portfolio report • Prepared for learning and interview discussion

1. Executive Summary

Customer churn occurs when a customer stops using a company's service. This project uses historical telecommunications customer data to explore patterns associated with churn, train a model that estimates churn probability, group customers by estimated risk, and present findings in a four-page Power BI dashboard.

The dataset contains 7,043 records: 1,869 customers labelled as churned and 5,174 labelled as not churned. The historical churn rate is 26.54%, while the retention rate is 73.46%. The selected model is a Random Forest classifier configured with balanced class weights.

Recorded test metric	Result	Plain-English interpretation
Accuracy	77.43%	Share of all test records classified correctly.
Churn precision	56.01%	Among customers predicted to churn, the share that actually churned.
Churn recall	69.79%	Among actual churners, the share identified by the model.
Churn F1-score	62.14%	A combined summary of precision and recall.
ROC-AUC	84.19%	How well the model ranks churners above non-churners across thresholds.

Interpretation: the model identifies a substantial share of churners, but it also generates false alerts. Metrics should be treated as preliminary because the test set was examined during earlier model comparisons and threshold analysis.

2. Project Title and Core Terms

Customer

A customer is a person or organization that uses a company's product or service. Each dataset row describes one telecom customer.

Churn

Churn means that a customer stops using a service or ends their relationship with the company. In this dataset, **Churn = Yes** marks a customer recorded as having left; **Churn = No** marks a customer recorded as not having left.

Retention

Retention means keeping customers and encouraging them to continue using the service. Retention actions may include resolving service problems, explaining plan options, improving support, or making relevant offers.

Analysis

Analysis means examining data to answer questions, compare groups, identify patterns, and support decisions.

In simple terms: this project studies why churn patterns occur in the available data, estimates which customers may be at risk, and helps a business prioritize possible retention actions.

3. Business Problem, Objectives and Scope

Business problem

A telecom company can lose recurring revenue when customers leave. If the company can recognize patterns associated with churn, it can investigate customer problems and direct retention resources toward customers who may need attention.

Objectives

- Understand the dataset and customer characteristics.
- Clean data and prepare it for analysis.
- Measure churn and retention and compare customer groups.
- Train and evaluate classification models.
- Estimate customer churn probabilities and create risk groups.
- Translate analysis into practical retention recommendations.
- Communicate the results through a Power BI dashboard.

Project scope

This is a portfolio analysis using historical data. It does not prove the causes of churn, guarantee future outcomes, or demonstrate that a retention campaign will work. Those claims would require additional evidence and testing.

4. Dataset and Data Dictionary

A **dataset** is an organized collection of records. A **row** represents one customer record, and a **column** stores one attribute. A **feature** is an input used by the model; the **target** is the outcome it predicts.

Dataset item	Value
Total customer records	7,043
Churn = Yes	1,869
Churn = No	5,174
Historical churn rate	26.54%
Historical retention rate	73.46%
Model input features	19
Training records in earlier run	5,634
Test records in earlier run	1,409

Column	Meaning
customerID	Identifier for an individual customer.
gender	Recorded gender category.
SeniorCitizen	Indicator for the dataset's senior-citizen category.
Partner / Dependents	Whether a partner or dependents are recorded.
tenure	Number of months the customer has stayed with the company.
PhoneService / MultipleLines	Phone-service subscription and multiple-line status.
InternetService	Recorded internet service type, such as DSL, fiber optic, or no internet.
OnlineSecurity / OnlineBackup	Recorded status of online security and online backup services.
DeviceProtection / TechSupport	Recorded device-protection and technical-support status.
StreamingTV / StreamingMovies	Recorded subscription status for streaming services.
Contract	Contract arrangement: month-to-month, one year, or two years.
PaperlessBilling	Whether paperless billing is recorded.
PaymentMethod	Recorded payment method.
MonthlyCharges	Recurring monthly amount charged.
TotalCharges	Accumulated charges recorded for the customer.
Churn	Historical target label: Yes or No.

5. Business Metrics Explained

Churn rate

The proportion of customers labelled as churned. Formula: churned customers ÷ total customers × 100. Here, 1,869 ÷ 7,043 × 100 = **26.54%**. This is the dataset-wide historical rate, not a live monthly rate.

Retention rate

The proportion labelled as not churned. Formula: retained customers ÷ total customers × 100. Here, 5,174 ÷ 7,043 × 100 = **73.46%**.

Tenure

Tenure is how long the customer has been with the company, measured in months.

Monthly charges and total charges

MonthlyCharges is the recurring monthly amount. TotalCharges is the accumulated amount recorded. A customer with short tenure may have relatively low total charges even when monthly charges are high.

Segmentation

Segmentation means dividing customers into meaningful groups. Here, customers are grouped into low, medium, and high estimated churn-risk categories.

6. Data Cleaning and Preprocessing

Data cleaning

Data cleaning means identifying and handling missing values, duplicates, inconsistent types, and invalid entries. In the notebook, TotalCharges was converted to a numeric value while the original text values were preserved separately.

Eleven blank TotalCharges values were associated with zero-tenure customers. They were filled with zero for analysis, while the original blank values were preserved in a separate column.

Target encoding

The target was encoded as 0 for Churn = No and 1 for Churn = Yes. This converts the two categories into numeric labels for model training.

Train-test split

An 80/20 split was used with stratification. The training set fits the model; the test set estimates how well the model predicts records it did not train on. Stratification helps preserve the class proportions in each split.

StandardScaler

StandardScaler transforms numeric values so that each feature is centered around its mean and scaled by its standard deviation.

One-hot encoding

OneHotEncoder converts categories into numeric indicator columns. For example, contract type can be represented by separate columns for month-to-month, one-year, and two-year contracts.

Pipeline

A pipeline combines preprocessing and modelling into one repeatable sequence, reducing the risk of applying different transformations during training and prediction.

7. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) means examining data before modelling to understand distributions, spot quality issues, and compare groups.

- Shorter-tenure customers showed higher observed churn.
- Month-to-month customers showed higher observed churn than customers on longer contracts.
- Electronic-check users showed higher observed churn.
- Churn patterns differed by internet-service type.
- Customers without online security or technical support showed higher observed churn.
- Monthly charges varied between churned and retained customers.

These observations are associations in this dataset, not proof that any single attribute causes churn. Other variables may be involved.

8. Machine Learning Concepts

Machine learning and classification

Machine learning uses patterns in historical data to make predictions. Classification predicts a category. Here, the categories are Churn and No Churn.

Class imbalance

Class imbalance means that outcome classes occur at different rates. Here, 5,174 customers did not churn and 1,869 did. This is why accuracy should be interpreted alongside precision, recall, and other metrics.

Random Forest

A Random Forest combines predictions from multiple decision trees. It can capture complex relationships and is less dependent on a single tree's decisions.

Balanced class weights

The model used `class_weight='balanced'`. This increases the relative importance of errors on the less frequent class during training.

Parameter	Value	Meaning
n_estimators	200	Number of decision trees in the forest.
max_depth	10	Maximum depth allowed for each tree.
min_samples_leaf	5	Minimum training records required in a leaf.
class_weight	balanced	Adjusts the penalty associated with each class.
random_state	42	Supports reproducibility of the setup.
n_jobs	-1	Uses available CPU cores for parallel processing.

In this project, “Balanced Random Forest” refers to a standard Scikit-learn RandomForestClassifier configured with balanced class weights; it is not necessarily the specialized BalancedRandomForestClassifier from imbalanced-learn.

Probability and decision threshold

The model estimates a churn probability. A decision threshold converts that score into a class. With threshold 0.55, a score of 0.55 or higher is labelled predicted churn. A lower threshold usually flags more customers; a higher threshold usually flags fewer. The threshold was chosen using cross-validation predictions to balance churn detection against false alerts.

9. Model Evaluation

Metric	Recorded result	Meaning
Accuracy	77.43%	Share of all test records predicted correctly.
Churn precision	56.01%	Share of predicted churners who actually churned.
Churn recall	69.79%	Share of actual churners identified by the model.
Churn F1-score	62.14%	Harmonic mean of precision and recall.
ROC-AUC	84.19%	Ranking ability across different classification thresholds.

Confusion matrix

A confusion matrix compares actual outcomes with predicted outcomes. In this project, churn is treated as the positive class.

Actual class	Predicted churn	Predicted no churn	Total
Churn	261 — True Positive	113 — False Negative	374
No churn	205 — False Positive	830 — True Negative	1,035
Total	466	943	1,409

- **True positive (261):** an actual churner correctly flagged as churn.
- **True negative (830):** a non-churner correctly classified as no churn.
- **False positive (205):** a non-churner incorrectly flagged as churn.
- **False negative (113):** an actual churner missed by the model.

Why these metrics matter

Accuracy summarizes all predictions, but it can hide poor performance on a less frequent class. Precision helps estimate how many retention contacts may be unnecessary. Recall indicates how many actual churners are detected. F1-score balances precision and recall. ROC-AUC assesses ranking ability across thresholds; it does not by itself determine the best business threshold.

10. Feature Importance

Feature importance estimates how useful each input was to the fitted Random Forest's predictive decisions. It does not prove cause and effect.

Rank	Feature	Importance
1	Tenure	0.1232
2	Month-to-month contract	0.1211
3	Total charges	0.0994
4	Two-year contract	0.0687
5	Monthly charges	0.0667
6	No online security	0.0587
7	No technical support	0.0492
8	Fiber-optic internet	0.0445
9	Electronic-check payment	0.0374
10	One-year contract	0.0247

Tenure and contract type were among the strongest predictive features in the recorded model. Importance values are specific to the fitted model and may be affected by relationships between features.

11. Customer Risk Segmentation

Risk categories were created from estimated churn probabilities: Low Risk below 0.30; Medium Risk from 0.30 to below 0.55; High Risk at or above 0.55.

Risk group	Customers	Dataset share	Average predicted probability	Observed churn rate
Low Risk	3,316	47.08%	11.2%	2.14%
Medium Risk	1,438	20.42%	42.6%	23.23%
High Risk	2,289	32.50%	74.5%	63.96%

Risk group	Suggested action
High Risk	Prioritize personalized outreach and investigate support, service, contract, or pricing concerns.
Medium Risk	Offer targeted support and review whether the current plan meets the customer's needs.
Low Risk	Maintain service quality and consider low-cost loyalty or satisfaction initiatives.

These risk-group summaries are retrospective: actual churn labels are available because this is historical data. In a live future prediction, actual outcomes would not yet be known.

12. Power BI Dashboard

Page 1 — Executive Overview

Purpose: summarize the customer base and headline churn/retention metrics. Core cards include total customers, churned customers, churn rate, retained customers, and retention rate. Charts compare churn across key customer categories.

Page 2 — Customer Churn Analysis

Purpose: explore how churn differs by tenure, internet service, technical support, online security, and monthly charges. Slicers allow users to filter the report by selected customer attributes.

Page 3 — Churn Risk & Retention

Purpose: connect observed customer patterns and risk groups to retention priorities. Risk-group counts and churn rates should be labelled clearly, and any summary table should not be presented as dynamically generated model predictions unless it is actually connected to the prediction output.

Page 4 — Model Performance

Purpose: present the recorded model metrics, confusion matrix, and decision threshold. Clearly distinguish actual outcomes from predicted outcomes. These are fixed evaluation results, not measures recalculated from the original customer table.

Common dashboard terms

KPI: Key Performance Indicator, a metric used to monitor an important outcome. **Card:** a visual showing one headline number. **Axis:** a chart scale or category direction. **Slicer:** an interactive filter that lets viewers focus on selected groups. **DAX:** Data Analysis Expressions, the formula language used to create measures and calculated columns in Power BI.

13. Business Recommendations

- Prioritize personalized retention outreach for customers in the high-risk group.
- Investigate service, support, pricing, and customer-experience concerns instead of assuming one factor explains churn.
- Review the experience of month-to-month customers and explain suitable plan options clearly.
- Check whether customers without online security or technical support need better service information or assistance.
- Estimate the monthly charges associated with flagged customers, but label this as exposure rather than guaranteed lost revenue.
- Use controlled experiments where practical to test whether retention offers actually improve retention.
- Monitor campaign costs, responses, and retained customers before scaling an intervention.

14. Project Deliverables

File	Purpose
README.md	Introduces the project, workflow, metrics, outputs, and limitations.
customer_churn_risk_predictions.csv	Customer-level predicted probabilities, labels, and risk categories.
model_evaluation_results.csv	Recorded evaluation metrics and confusion-matrix values.
customer_risk_summary.csv	Counts and observed outcomes by risk category.
retention_strategy.csv	Suggested actions by risk group.
customer_churn_project_summary.csv	High-level summary of the project.
customer_churn_random_forest.pkl	Saved preprocessing and model pipeline for reuse.
Power BI report	Interactive dashboard for customer analysis and model results.

The saved model pipeline expects compatible input columns and data types. Customer-level risk files contain sensitive-granularity information and should not be published publicly without appropriate review.

15. Limitations and Next Improvements

- The dataset may not reflect every telecom company or current market conditions.
- Predicted probabilities are estimates and are not guarantees.
- Feature importance and observed group differences do not prove causation.
- The earlier test set was examined during model comparisons and threshold analysis; the recorded metrics are not a completely untouched final evaluation.
- The current precision means some flagged customers did not churn in the test set, so outreach costs matter.
- A stronger next version should preserve a new untouched test set, tune models and thresholds using training-only cross-validation, and evaluate once at the end.

- Additional business value could come from SQL-based KPI validation, estimated monthly charges at risk, and measured retention campaign outcomes.

16. Glossary

Term	Plain-English definition
Attribute / variable	A recorded property, such as tenure or payment method.
Business intelligence (BI)	Using data reports and analysis to support decisions.
Categorical feature	An input containing labels or groups rather than a continuous number.
Classification	Predicting a category such as Churn or No Churn.
Cross-validation	Repeatedly fitting and validating on different training-data splits to compare choices.
Data cleaning	Finding and handling missing, duplicate, inconsistent, or invalid data.
EDA	Exploratory Data Analysis: inspecting data to understand patterns.
Feature	An input used by the predictive model.
Feature importance	A model-specific estimate of an input's predictive contribution.
False negative	An actual churner predicted as no churn.
False positive	A non-churner predicted as churn.
F1-score	A combined measure of precision and recall.
KPI	Key Performance Indicator: a metric used to monitor performance.
Pipeline	A repeatable sequence of data transformations and model steps.
One-hot encoding	Turning categories into numeric indicator columns.
Precision	Among predicted positives, the share that are actually positive.
Preprocessing	Transforming data into a format suitable for modelling.
Probability	A score representing the model's estimated likelihood of an outcome.
Decision threshold	The cutoff used to turn a probability score into a predicted class.
Recall	Among actual positives, the share identified by the model.
ROC-AUC	A measure of how well the model ranks positives above negatives across thresholds.
Target	The outcome the model is trained to predict.
Test set	Records reserved to evaluate a model after fitting.
Training set	Records used to fit a model.
True negative	An actual non-churner predicted as no churn.
True positive	An actual churner predicted as churn.

17. Interview-Ready Project Explanation

“I developed a Customer Churn & Retention Analysis project using the IBM Telco Customer Churn dataset, containing 7,043 customer records. I used Python and Pandas for cleaning and exploratory analysis, then built a Scikit-learn pipeline with numerical scaling, one-hot encoding, and a class-weighted Random Forest classifier. I evaluated the model using accuracy, precision, recall, F1-score, ROC-AUC, and a confusion matrix. I then grouped customers by estimated churn risk and created a four-page Power BI dashboard to communicate churn patterns, model performance, and retention recommendations. The recorded model achieved 69.79% churn recall and 84.19% ROC-AUC, with the limitation that the test set had been examined during earlier model comparisons.”

Questions to practise

- What is churn, and why does it matter to a telecom company?
- How are churn rate and retention rate different?
- Why are categorical features one-hot encoded?

- What does class_weight='balanced' do?
- How do precision and recall differ?
- What do the four confusion-matrix values mean?
- Why was a threshold of 0.55 selected?
- Why does feature importance not prove causation?
- How would you test whether a retention campaign works?

18. Conclusion

This project brings together data cleaning, exploratory analysis, machine learning, risk segmentation, business recommendations, and Power BI reporting. It demonstrates a practical analytical workflow while making clear that model predictions are uncertain and must be evaluated carefully.

The most important future improvement is a clean final evaluation on data that has not been used to compare models or choose thresholds. Beyond that, SQL validation and quantified retention economics would make the project more useful for a business and more compelling in a Data Analyst interview.

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